

Robotics system design, from level 1 to the top

This folder teaches you to design a robot's software as a whole, not just to write one algorithm. It starts with no assumptions: level 1 has no code and no jargon. Each level builds on the one before. Read them in order.

Level	File	What you can do after it
0	The tree	The whole subject as one tree; redraw it weekly from memory
1	What a system is	Draw any robot as boxes and arrows, label the arrows, name a trade-off
2	The robot software stack	Name every box in a real robot's stack, explain topics/services/actions, frames and time
3	Timing, reliability and safety	Build a latency budget, handle failures, draw a state machine, test a robot system
4	Fleets, cloud and scale	Design for 200 robots: what runs where, traffic, updates, monitoring, safety cases
5	Design questions, worked	Take "design a robot that does X" to a full design in 40 minutes, six worked examples

How to study this

- One level per sitting. Draw every diagram yourself on paper before reading the one in the text. Drawing is the skill; reading is not.
- Do the "Try it" exercises. Level 5 contains the answers to the level 1 exercises.
- After level 3, go back to the coding tutorial and notice which box each algorithm lives in: BFS and A* in [tutorial/03](#) are the planner box, the Kalman filter and PID in [tutorial/06](#) are the state estimation and control boxes.

The one idea

Every robot is **sense → think → act**. Designing is drawing those boxes, labelling what flows between them and how often, and choosing trade-offs on purpose. The hard problems live on the arrows: how fresh the data is, how much of it there is, and what happens when it stops.